

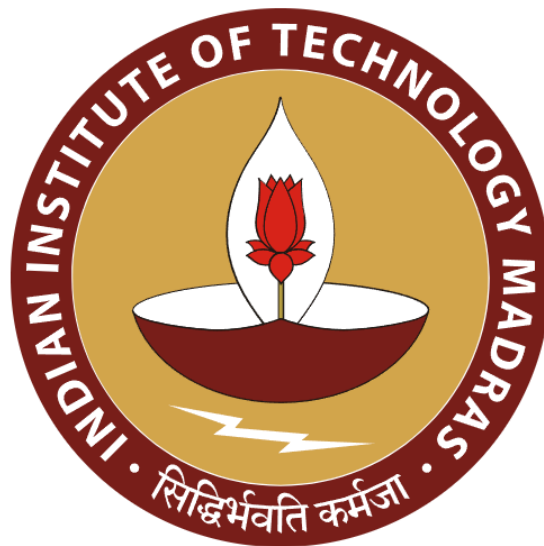
CASE STUDY OF A KIRANA STORE:PROBLEMS,STRATEGIES AND APPROACH

Mid term submission report for the BDM capstone Project

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1.Executive Summary and Title:

This report presents the mid-term findings for the Business Data Management capstone project, which investigates the revenue decline experienced by Kanish Stores, a local grocery store. Kanish Stores face challenges such as inconsistencies between daily purchases and sales, and excessive inventory of low-demand items.

The primary objective of this project is to analyze the store's transaction data to identify areas of inefficiency in sales, and inventory management, and propose solutions to enhance profitability.

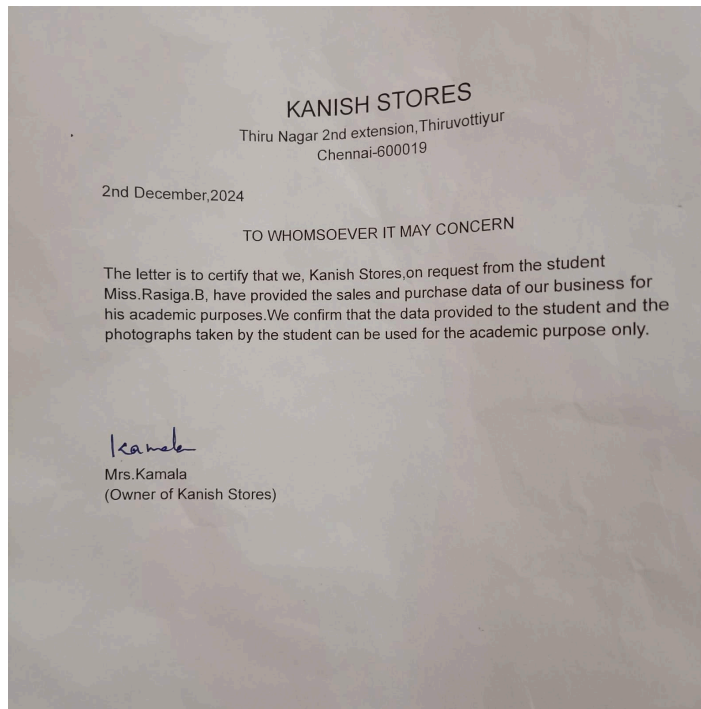
This expected outcome of this project aims to assist the store in enhancing its performance. By analyzing data, we will identify areas for improvement, such as optimizing inventory to meet customer demand, boosting daily sales, reducing reliance on credit sales, and improving cash flow and profitability.

In summary, this mid term report offers a preliminary analysis of the business data and provides an idea of the root cause of the challenges faced by the store.

2.Proof of Originality:

The data is collected from the business with the consent from the business to use it for academic purposes only. Further photos of the shop maintained by the business were attached with this document.

Communications with the business were mostly over meetings. Initial conversations with the business is attached in the document



2a. Consent letter from the organization



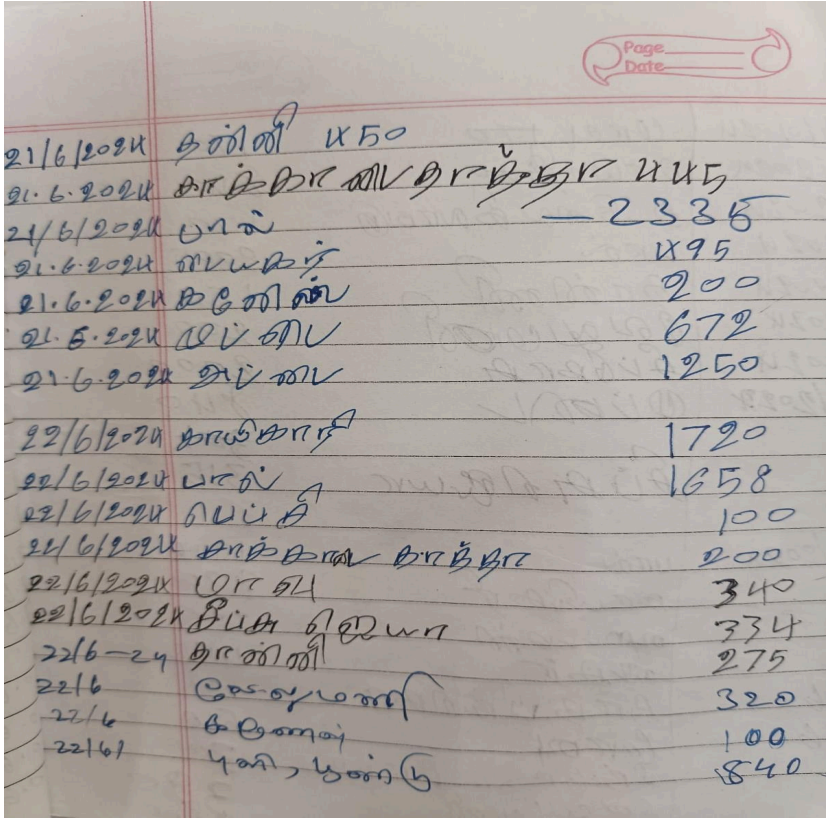
2b. Image of the Shop

Meeting with owner:  video link.mp4

3 Data Collection and Preprocessing:

The data required for analysis is collected for a period of 3 months from the business. All transaction information was manually recorded in a notebook because the store lacked an official ledger. The handwritten entries from the notebook were manually digitized into Google Sheets, making the raw data accessible for analysis. The data has been aggregated on a weekly basis to provide a clear understanding of sales trends and performance over time.

The data had numerous redundant rows with the same date and timestamp. Those rows were deleted for better identification of unique transactions.



21/6/2024	മുതിർത്തു	150
21.6.2024	മുതിർത്തു മുതിർത്തു	145
21/6/2024	പുതിയ	2335
21.6.2024	മുതിർത്തു	195
21.6.2024	മുതിർത്തു	200
21.6.2024	മുതിർത്തു	672
21.6.2024	മുതിർത്തു	1250
22/6/2024	മുതിർത്തു	1720
22/6/2024	പുതിയ	1658
22/6/2024	മുതിർത്തു	100
22/6/2024	മുതിർത്തു മുതിർത്തു	200
22/6/2024	മുതിർത്തു	340
22/6/2024	മുതിർത്തു	334
22/6-24	മുതിർത്തു	275
22/6	മുതിർത്തു	320
22/6	മുതിർത്തു	100
22/6	മുതിർത്തു	1840

3a.Information maintained by the shop owner manually in notebook

A	B	C	D	E	F	G	H	I	J
Date	Item Name	Category	Price	Quantity	Total Cost	Measurement	Profit/Loss	Transaction Type	
2023-07-01	Urad Dal	Grocery	50	4	200	40	40	Cash	
2023-07-01	Sugar	Grocery	40	1	40	35	5	Credit	
2023-07-01	Oil	Grocery	150	3	450	135	45	Credit	
2023-07-01	Ghee	Grocery	500	1	500	450	50	Credit	
2023-07-01	Milk & Dai	Grocery	50	1	50	40	10	Cash	
2023-07-01	Dry Fruits	Grocery	600	3	1800	550	150	Cash	
2023-07-01	School Bag	Non-Groce	500	1	1000	600	-200	Credit	
2023-07-01	Soap Disp	Non-Groce	150	1	150	250	-100	Credit	
2023-07-01	Umbrella	Non-Groce	250	1	1000	350	-400	Cash	
2023-07-02	Urad Dal	Grocery	50	4	200	40	40	Credit	
2023-07-02	Sugar	Grocery	40	3	120	35	15	Cash	

3.b Format of the Data in Excel format

Further the data from spreadsheets are updated to Google Colab for better interpretation and visualization

Link to data : [updatedfinalmid.xlsx](#)

Colab Link : [Untitled2.ipynb](#)

4.Data Description And Metadata:

Two categories of data such as sales and purchase were collected from the business.

4.1 Sales:

Sales transactions represent the store's revenue, reflecting the goods sold to customers or other retail shops.

4.2 Purchase:

Purchase transactions track the procurement of goods from wholesale stores which are essential for the store's daily operations. These contribute to the store's inventory and impact its sales and profitability.

Metadata of the Columns:

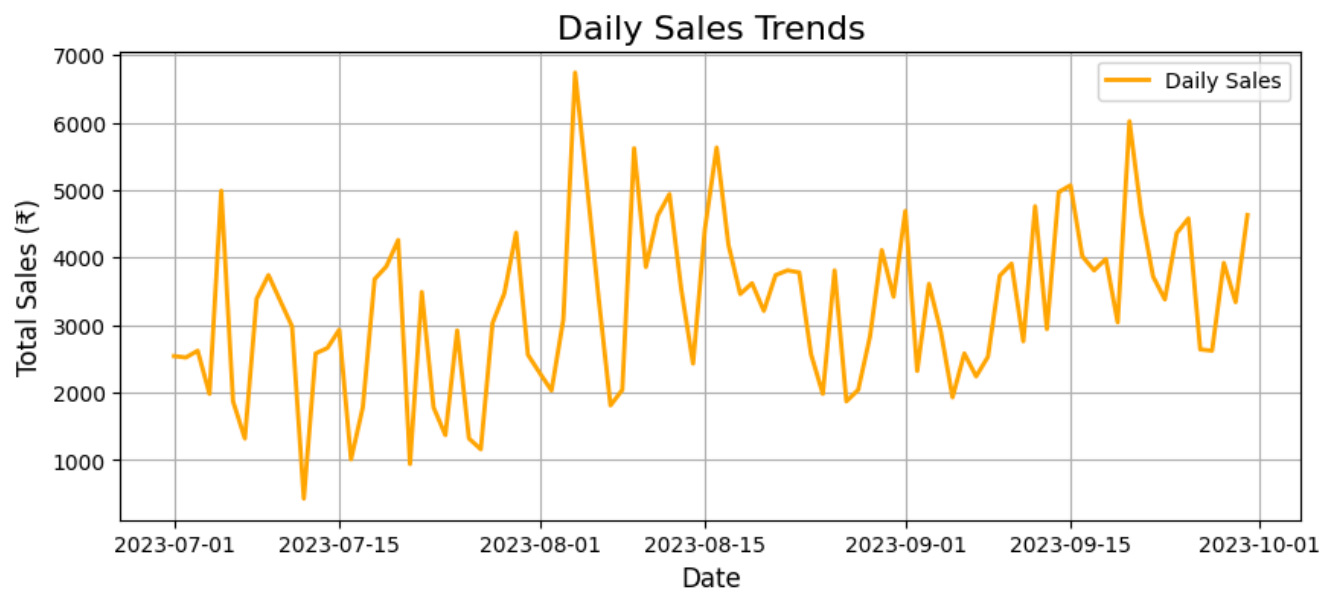
- 1.Date: The date on which that particular purchase/sales transaction was made
- 2.Item Name: The name of the item that was sold or procured.
- 3.Category: Specifies whether the product is a grocery or non-grocery item.
- 4.Price: The selling price of the product per unit.
- 5.Quantity: The number of units sold or procured.
- 6.Total Cost: The total value of the transaction, calculated as $\text{Price} \times \text{Quantity}$.
- 7.Procurement Cost: The cost incurred for procuring the goods.
- 8.Profit/Loss: The difference between the selling price and the procurement cost.
- 9.Transaction Type: Specifies whether the transaction was conducted in "cash" or on "credit."

5.Detailed Explanation of Analysis Process:

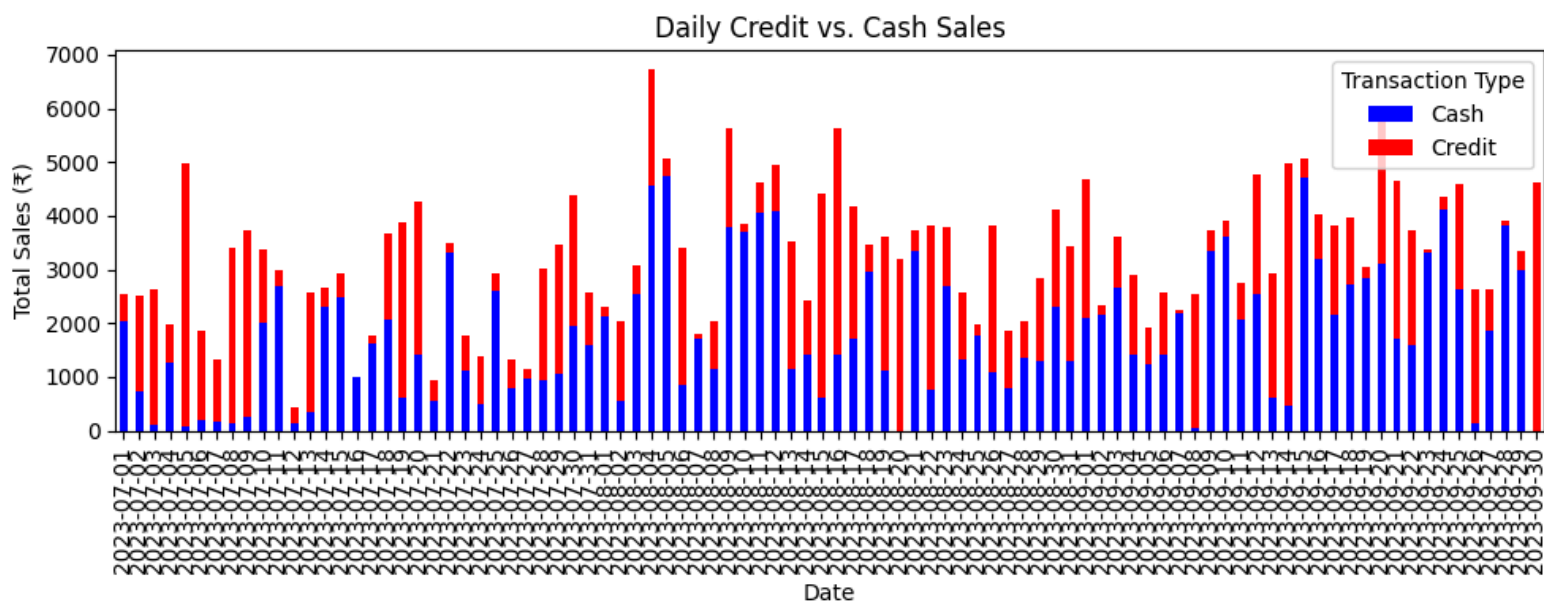
The data analysis for this business data should revolve around the problems faced by the business. We are trying to analyze 3 major aspects from the data

1. Total value of sales and the trend of the sales value for a period of 3 months
2. Trend Analysis of Credit vs Cash Sales Over Time
3. Analysis of Profitable vs Loss Contributing items

Total Sales Vs Date



5a. Line chart to show the trend of sales over a period of one year



5b. stacked bar chart to visualize the daily split between credit and cash sales.



5c. Bar chart to visualize the profit and loss data.

6.Results And Insights:

During the analysis process, the following insights were observed

1.Cash sales(sales between the business and the customer directly where the customer is the end user of the product) contributes 54.38% of the total sales.

2.There appears to be no significant relation between the date and cost of goods sold to customers. Analyzing the total sales value across different months would provide a clearer insight into the trends in sales over time.

3.Profit contributing items are dry fruits,dairy products and ghee(Grocery) whereas loss contributing items are umbrellas,school bags and soap dispensers(Non Grocery)

4.A 54% gross profit margin means that for every rupee of sales, ₹0.54 is the profit before accounting for other business expenses, with the cost of goods sold representing ₹0.46 of the total revenue.

5.The grocery category has earned a profit of Rs.31,340 whereas the Non-Grocery category has incurred a loss of Rs. 67,200 .This indicates that the store is spending more on procuring non-grocery items than it earns from selling them.

7.Expected Timeline:

- The final report for the analysis is planned to be completed by Jan 17th, 2024.
- The final report completion and result analysis is targeted for Jan 19th, 2023.
- The final report will serve as an analysis of income generated, profit margin, and consumer demand shifts - aiding the company to anticipate future needs through data-driven forecasting and decision-making.

